

System Schematics & Architecture

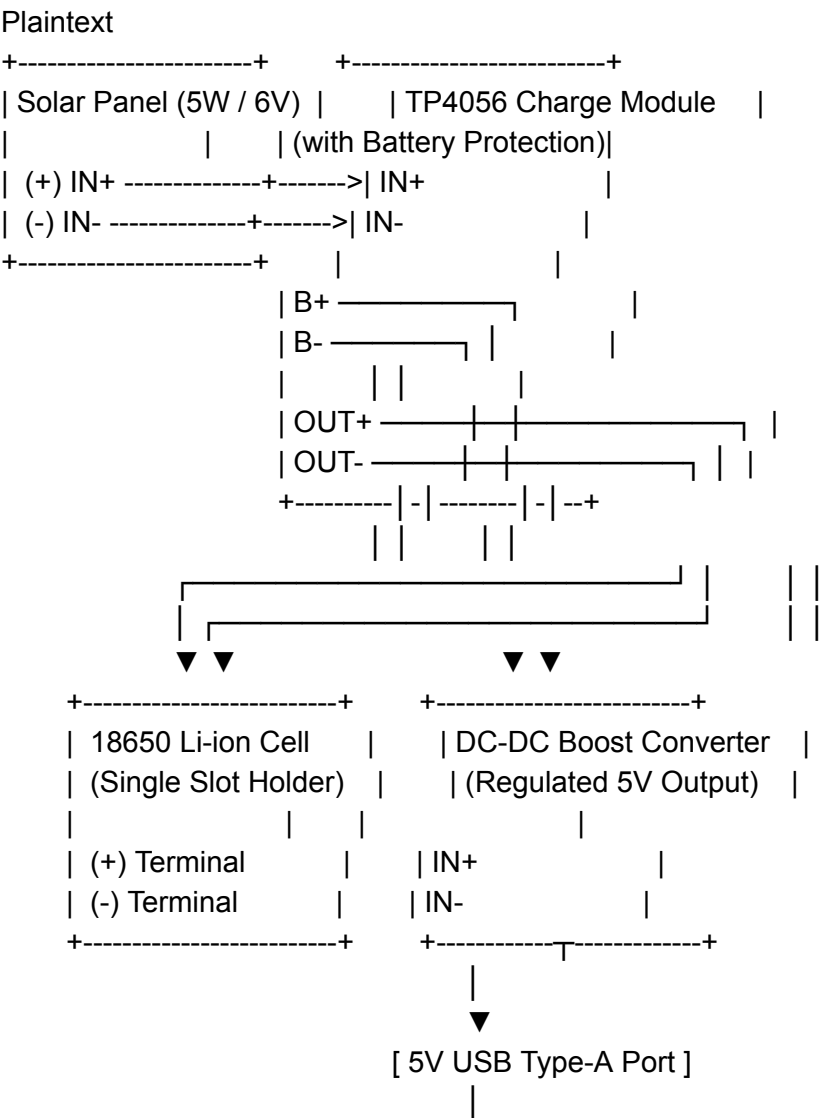
This document details the hardware architecture and electrical interconnects for all three evolution phases of the **Smart Solar Tracker & Off-Grid Power Hub**.

Phase 1: Pocket MVP (Static 5W Solar Charger)

Overview

A baseline low-power system designed to validate PV energy collection, single-cell Lithium-ion battery charging, and 5V regulated output.

Block Diagram



▼
Target Device / Load

Pinout & Wiring Specification

Source Component	Source Pin	Target Component	Target Pin	Wire / Function
Solar Panel (5W, 6V)	VCC (+)	TP4056 Module	IN+	Solar Power Input (+)
Solar Panel (5W, 6V)	GND (-)	TP4056 Module	IN-	Solar Power Return (-)
TP4056 Module	B+	18650 Battery Holder	Positive Terminal (+)	Battery Charging (+)
TP4056 Module	B-	18650 Battery Holder	Negative Terminal (-)	Battery Charging (-)
TP4056 Module	OUT+	DC-DC Boost Module	IN+	Protected Load Power (+)
TP4056 Module	OUT-	DC-DC Boost Module	IN-	Protected Load Return (-)

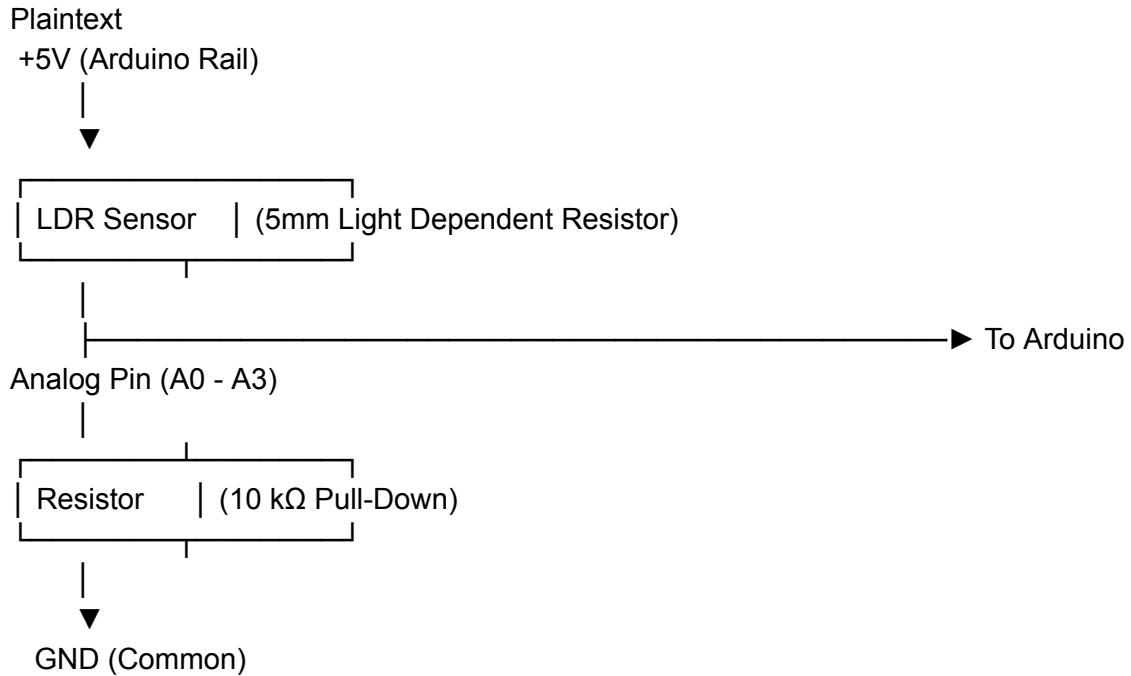
Phase 2: Active Dual-Axis Solar Tracker (Arduino Automation)

Overview

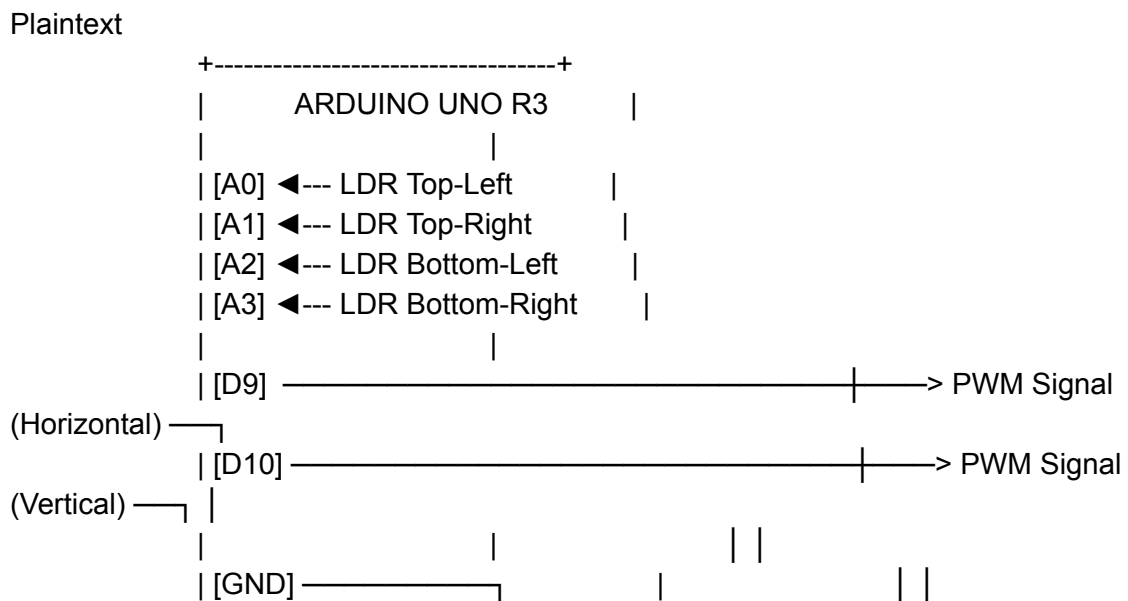
A closed-loop control subsystem combining a 4-quadrant Light Dependent Resistor (LDR) sensor matrix with dual servo actuators (horizontal/azimuth and vertical/elevation) managed by an Arduino microcontroller.

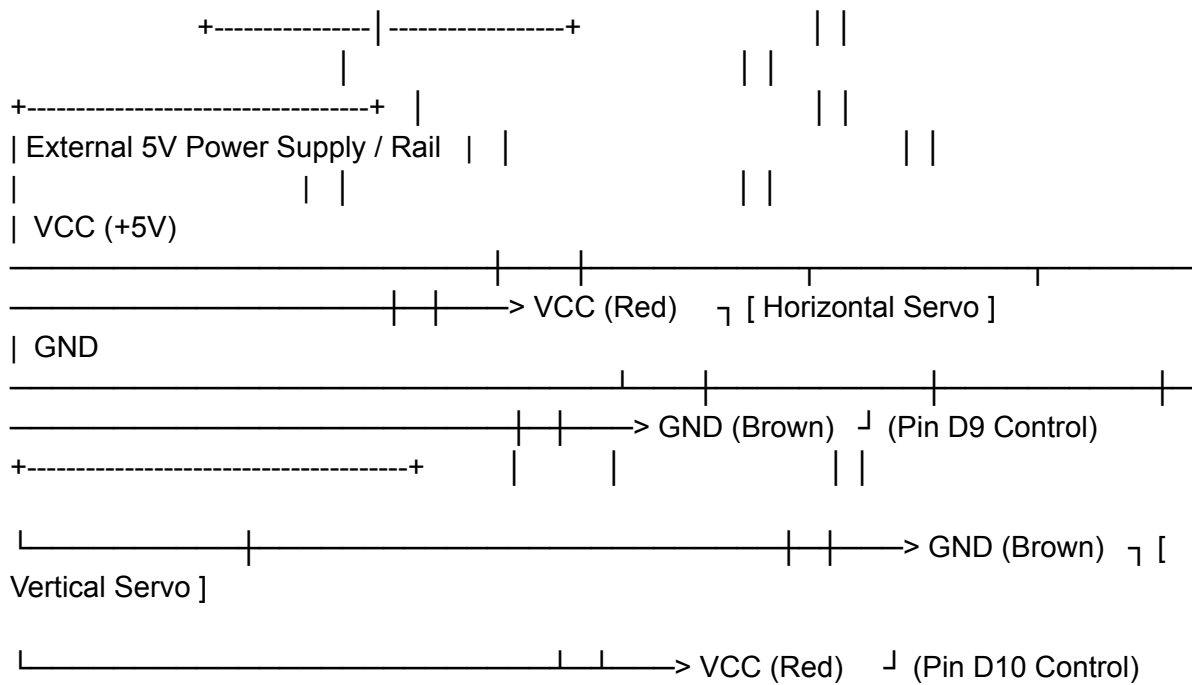
1. Sensor Conditioning Circuit (Voltage Divider Matrix)

Each of the four LDR sensors uses a pull-down resistor to convert light-dependent resistance changes into an analog voltage input range readable by the ADC.



2. Microcontroller & Actuator Interconnects





IO Mapping Table

Component	Signal / Pin	Arduino Pin	Description
LDR Sensor 1	Top-Left Analog Out	A0	Upper-left quadrant light sensing
LDR Sensor 2	Top-Right Analog Out	A1	Upper-right quadrant light sensing
LDR Sensor 3	Bottom-Left Analog Out	A2	Lower-left quadrant light sensing
LDR Sensor 4	Bottom-Right Analog Out	A3	Lower-right quadrant light sensing
Horizontal Servo	PWM Control Line	D9	Azimuth axis movement
Vertical Servo	PWM Control Line	D10	Elevation axis movement

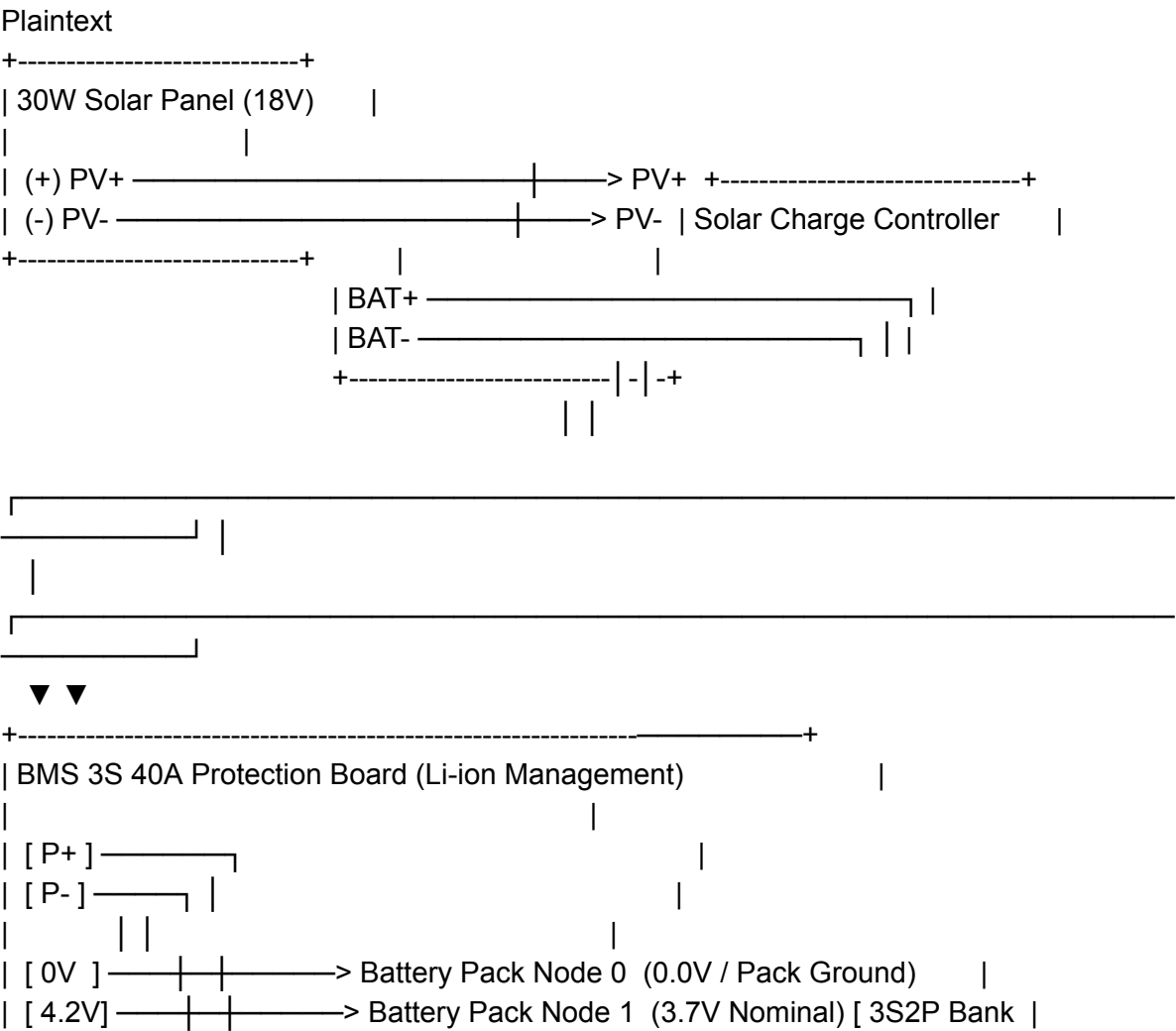
Servos Rail	VCC / GND	External 5V / Shared GND	Dedicated power rail to prevent MCU brownouts
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Phase 3: High-Capacity Off-Grid Power Hub (3S2P System)

Overview

A high-power energy management system integrating a 30W PV panel, a 3S2P Lithium-ion battery bank (11.1V nominal / 12.6V peak), a 3S 40A BMS protection board, and a multi-port step-down Buck converter providing fast-charging protocols (USB-C PD & QC3.0).

System Architecture Schematics



Solar Input Power	30W Max (18V Vmp)	Optimized for high-yield collection
BMS Continuous Discharge	40A (Max)	Over-charge, over-discharge, short-circuit protected
Output Capabilities	Dual USB-C PD + Single USB-A QC3.0	Capable of driving laptops and fast-charge peripherals